Shuyi (Ryan) Chen

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CONTACT

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EDUCATION

Carnegie Mellon University, Heinz College — Pittsburgh, PA
Ph.D. in Information Systems Management — 2024 – Summer 2027 (*expected*)
Advisor: Shixiang (Woody) Zhu
Master of Information Systems Management (Highest Distinction) — 2021 – 2023

The Chinese University of Hong Kong, Shenzhen — Shenzhen, China
B.B.A. (Hons.) in Global Supply Chain and Logistics Management; Minor in Statistics — 2017 – 2021
First Class Honours
Visiting Student, University of California, Berkeley — 2018

RESEARCH INTERESTS

Methodology: decision-focused learning; robust, chance-constrained, and data-driven optimization; conformal prediction and uncertainty quantification; generative and time-series modeling.
Applications: power-system resilience to wildfires and extreme weather; energy storage and grid operations; trustworthy machine learning.

PUBLICATIONS

*Equal contribution.

JOURNAL ARTICLES

[J1] **Global-Decision-Focused Neural ODEs for Proactive Grid Resilience Management** [arXiv] [code]
Shuyi Chen, Ferdinando Fioretto, Feng Qiu, Shixiang Zhu.
IEEE Transactions on Smart Grid.

REFEREED CONFERENCE PROCEEDINGS

[C1] **Learning Polyhedral Conformal Sets for Contextual Robust Optimization** [arXiv]
Shuyi Chen*, Wenbin Zhou*, Shixiang Zhu.
Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C2] **Gen-DFL: Decision-Focused Generative Learning for Robust Decision Making** [arXiv]
Prince Zizhuang Wang, Shuyi Chen, Jinhao Liang, Ferdinando Fioretto, Shixiang Zhu.
International Conference on Learning Representations (ICLR), 2026.
IISE DAIS Track Best Paper Award.

[C3] **Counterfactual Fairness through Transforming Data Orthogonal to Bias** [arXiv] [code]
Shuyi Chen, Shixiang Zhu.
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Research Track, 2025.

[C4] **Uncertainty-Aware Robust Learning on Noisy Graphs** [arXiv] [poster] [code]
Shuyi Chen, Kaize Ding, Shixiang Zhu.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025.
Short version at the NeurIPS 2023 Workshop on New Frontiers in Graph Learning.

[C5] **API-Assisted Code Generation for Question Answering on Varied Table Structures** [arXiv]
Yihan Cao*, Shuyi Chen*, Ryan Liu*, Zhiruo Wang, Daniel Fried.
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

Working Papers

Under Revision

[W1] **Enhancing Electricity-System Resilience with Adaptive Robust Optimization and Conformal Uncertainty Characterization** [arXiv] [media]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
Minor revision, IEEE Open Access Journal of Power and Energy.
INFORMS ENRE Best Student Paper Award.

In Preparation

[W2] **Large-Scale Resilience Planning for Wildfire-Prone Electricity Systems via Adaptive Robust Optimization** [arXiv] [poster] [demo]
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
To be submitted to European Journal of Operational Research.

[W3] **The Price of Looking Ahead: Decision-Focused Learning for Model Predictive Control in Energy Storage**
Shuyi Chen, Shuai Wang, Shixiang Zhu.
In preparation for conference submission. Earlier version: "Balancing Planning Myopia and Forecast Error: Decision-Focused Sequential Learning."
INFORMS DMDA Workshop Best Paper Finalist (Applied Track).

[W4] **Rare-Chance-Constrained Resilience Planning for Power Systems under Wildfires and Cascading Hazards**
Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi.
In preparation for journal submission.

Honors & Awards

INFORMS DMDA Workshop Best Paper Finalist (Applied Track)	2026
Young Researchers Workshop participant, Cornell University	2026
IISE DAIS Track Best Paper Award	2026
INFORMS ENRE Best Student Paper Award	2025
KDD Student Travel Award; NSF Travel Grant (IMSI Workshop)	2025
INFORMS Workshop on Data Science Student Scholarship	2024
PwC Technology Impact Office / CMU Digital Transformation Center Research Fellowship	2023
CUHK Presidential Award (top 10 of 1,200 graduates)	2021

Teaching

Lecturer, *A/B Testing* (95-819), Heinz College, Carnegie Mellon University 2026
Teaching interests: optimization and operations research; machine learning and data analytics; A/B testing and causal inference; energy and sustainability analytics.

Presentations

HOPE Research Workshop, University of Notre Dame (invited talk) 2026
Large-scale resilience planning for wildfire-prone electricity systems
ACM SIGKDD Conference, Toronto 2025
Counterfactual fairness through transforming data orthogonal to bias
NSF/IMSI Workshop on UQ and ML for Complex Physical Systems, Chicago (poster) 2025
Electricity-system resilience with adaptive robust optimization and conformal uncertainty
INFORMS Workshops on Data Science and on Data Mining & Decision Analytics 2024

Counterfactual fairness; decision-focused neural ODEs for grid resilience

Research Experience

Argonne National Laboratory — Lemont, IL
Researcher — 2025

- Developed decision-focused neural ODEs for proactive grid resilience, reducing decision regret by up to **75%** versus two-stage baselines [J1].
- Led the 2025 INFORMS DMDA Workshop Data Challenge on extreme-weather power-outage prediction, which drew **54 teams from 31 institutions**.

Industry Experience

ByteDance — Beijing, China
Data Scientist — 2023 – 2024

- Built Prophet-based anomaly detection with statistical residual tests (**3% MAPE**), flagging three major business anomalies for timely mitigation.
- Applied causal inference to Local Services growth interventions, contributing to a **2% increase** in overall user activity.

Professional Service

Competition Co-Chair: INFORMS DMDA Workshop Data Challenge (2025); IISE ESD Energy Analytics Challenge (2026).
Organizing Committee: YinzOR, Speakers and Talks Committee (2026).
Session Chair: INFORMS Energy Systems session (2025).
Journal Reviewer: INFORMS Journal on Data Science; Journal of Agricultural, Biological, and Environmental Statistics.
Conference Reviewer: NeurIPS; ICML; ICLR; KDD; AISTATS; ICASSP; IEEE PES General Meeting (2023 – 2026).

References

Shixiang (Woody) Zhu (Advisor)
Assistant Professor
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Ramteen Sioshansi
Professor, Engineering and Public Policy
Carnegie Mellon University
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Feng Qiu
Principal Computational Scientist
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Ferdinando (Nando) Fioretto
Associate Professor, Computer Science
University of Virginia
fioretto@virginia.edu